

Shapes survey dataset requires structural alignment between table metadata and image files before training an image classifier.

The shapes_survey.zip dataset contains 38 analyzed images comprising 16 circle images, 12 square images, and 10 triangle images, with an average data quality score of 92.4 out of 100 before cleaning and 97.4 out of 100 after cleaning. While the image data scored 99.7 out of 100 with zero corrupt files or visual mislabels, the table metadata scored 95.1 out of 100 and introduced critical label disagreements, missing files, and version mismatches. Specifically, there are 3 table rows pointing to missing images, 4 unreferenced images, and 3 instances of label disagreements between the table and folder structure.

Data quality before	After cleaning	Items	Numbers fact-checked
92.4/100	97.4/100	76 → 75	8

Cross-type findings

WARNING Table rows point to missing images

There are 3 table rows that reference image files not present in the dataset.

Recommendation: Remove these rows from the table or locate the missing files to ensure data integrity.

Evidence: mix_links_001

WARNING Unreferenced images exist in the dataset

The dataset contains 4 image files that are not described by any row in the table.

Recommendation: Verify if these images should be included and add corresponding entries to the table.

Evidence: mix_links_001

CRITICAL Label disagreements between table and folders

There are 3 instances where the label in the table contradicts the folder structure of the image.

Recommendation: Manually audit these 3 images to determine the correct label and update the table or move the files.

Evidence: mix_links_001

INFO Dataset quality bottleneck

The overall dataset quality is limited by the table, which has a quality score of 95.1 out of 100, compared to the image quality of 99.7 out of 100.

Recommendation: Focus future cleaning efforts on the table metadata to align its quality with the image assets.

Evidence: mix_overview_001

WARNING Multiple table rows reference single image

The evidence shows 1 file is referenced by more than one row in the table.

Recommendation: Review the table annotations to remove duplicate row references pointing to the same image file.

Evidence: mix_links_001

WARNING Table row count version mismatch

The table has 37 rows after cleaning, but the link analysis refers to 38 rows, suggesting a mismatch in the version of the table being analyzed.

Recommendation: Re-run the link analysis pipeline using the updated table version containing 37 rows.

Evidence: mix_overview_001, mix_links_001

Each data type

Each type was profiled, cleaned, analyzed, and reviewed by its own crew. Their full reports are in cleaned_by_type.zip.

Table · 1 file(s) · quality 85.1 → 95.1 · rows 38 → 37

- Exact duplicate row removed: The raw dataset contained 1 exact duplicate row representing 2.63% of the data, which was successfully removed during cleaning.
- Data quality score improved: The overall data quality score increased from 85.1 out of 100 to 95.1 out of 100 after handling missing values, hidden-missing tokens, and duplicate rows.
- Missing values and hidden-missing tokens handled: Columns confidence and area_px each had missing values and hidden-missing tokens accounting for 2.63% of the raw data, which were imputed using median imputation.
- High missingness in notes column: The notes column had 44.74% missing values and hidden-missing tokens in the raw data, preserving a missingness indicator during cleaning.

Cleaning steps

1. standardize_null_tokens: Changed 1 value in 'confidence', all now missing (e.g. 'N/A' → (missing)); changed 1 value in 'area_px', all now missing (e.g. 'N/A' → (missing)); changed 17 values in 'notes', all now missing (e.g. '' → (missing); 'N/A' → (missing)).
2. trim_whitespace: Nothing needed changing: the data already met this rule.
3. drop_exact_duplicates: Removed 1 row.
4. mark_as_id: Nothing needed changing: the data already met this rule.
5. parse_percent: Converted 'confidence' to numbers; changed 36 values in 'confidence' (e.g. '70%' → 70; '95%' → 95; '96%' → 96).
6. cast_numeric: Converted 'area_px' to numbers.
7. add_missing_indicator: Marked 1 row in a new 'confidence_was_missing' column; marked 1 row in a new 'area_px_was_missing' column.
8. impute_median: Changed 1 value in 'confidence' (e.g. (missing) → 86).
9. impute_median: Changed 1 value in 'area_px' (e.g. (missing) → 4966).
10. add_missing_indicator: Marked 16 rows in a new 'notes_was_missing' column.
11. impute_constant: Changed 16 values in 'notes' (e.g. (missing) → 'none').
12. standardize_column_names: Nothing needed changing: the data already met this rule.

Image · 38 file(s) · quality 99.7 → 99.7 · images 38 → 38

- Dataset size of 38 images: The dataset is based on a sample of 38 analyzed images, comprising 16 circle images, 12 square images, and 10 triangle images.
- Class imbalance across shape categories: Circles represent 42.11 percent of the dataset with 16 count, while squares represent 31.58 percent with 12 count, and triangles represent 26.32 percent with 10 count, resulting in an imbalance ratio of 1.6.
- Absence of corrupt or low-quality images: The dataset contains 0 corrupt images and 0 blurry images out of 38 analyzed files, yielding a clean data score of 99.7 out of 100.
- Zero duplicates and cross-class leakage: There are 0 exact duplicate groups, 0 near-duplicate groups, and 0 files duplicated across classes in the dataset of 38 analyzed images.
- No visual mislabels identified by AI review: Based on a vision review AI observation, there are 0 suspected mislabeled images out of 38 analyzed files across all categories.
- Cleaning process preserved all images: The cleaning pipeline processed 38 images and removed 0 files, moving from 38 initial images to 38 clean images.

Cleaning steps

1. remove_corrupt: Nothing needed changing: the data already met this rule.
2. drop_exact_duplicates: Nothing needed changing: the data already met this rule.
3. convert_to_rgb: Marked 38 images: to rgb.

Table-to-file links

annotations.csv, column 'image': 35 of 38 rows match a file; 3 point to missing files; 4 of 38 files have no row; column 'label' disagrees with the folder for 3 rows.

Problem	Example
Row names a missing file	circle_100.jpg
Row names a missing file	circle_101.jpg
Row names a missing file	circle_102.jpg
File with no row	shapes_survey/images/circles/circle_003.jpg
File with no row	shapes_survey/images/squares/square_012.jpg
File with no row	shapes_survey/images/triangles/triangle_005.jpg
File with no row	shapes_survey/images/triangles/triangle_010.jpg
Label disagrees (row 4)	shapes_survey/images/circles/circle_004.jpg: table 'square', folder 'circles'
Label disagrees (row 14)	shapes_survey/images/circles/circle_014.jpg: table 'triangle', folder 'circles'
Label disagrees (row 24)	shapes_survey/images/squares/square_008.jpg: table 'circle', folder 'squares'

Next steps

1. Resolve the 3 instances of label disagreements between the table metadata and the folder structure.
2. Investigate and fix the 3 table rows that reference missing image files and the 4 unreferenced images.
3. Address the table row count version mismatch where the cleaned table has 37 rows while link analysis refers to 38 rows.
4. Handle the class imbalance across shape categories, which has an imbalance ratio of 1.6.
5. Proceed with image model training once metadata alignment issues are fully resolved.

How the agents checked themselves

Model calls 24 · guardrail catches 8 · self-corrections 4 · review rounds 4 · model switches 7

+7s Switched models for the strategist

gemini-3.8-flash hit its daily limit; trying the next model in the pool.

+12s Switched models for the strategist

gemini-3.7-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

+18s Switched models for the strategist

gemini-3.6-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

+20s Switched models for the strategist

gemini-3.5-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

+22s Cleaning plan failed the dry run

Step 8 (impute_median on ['area_px']) failed: TypeError: Cannot perform reduction 'median' with string dtype. The column is still text: convert it first (cast_numeric, strip_currency, or parse_percent).

+26s Revised output accepted

+27s Findings failed the fact check

Finding 1 ('Small dataset size of 38 images'): 'classes.circles.count' isn't in ['img_overview_001', 'img_overview_002']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them. Finding 3 ('Absence of corrupt or low-quality images'): the statement uses 38...

+31s Findings failed the fact check

Finding 3 ('Class distribution imbalance'): 'label.top' isn't in ['tbl_columns_002']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them.

+31s Findings failed the fact check

Finding 6 ('Cleaning process preserved all images'): 'images.before' isn't in ['cleaning_summary_001']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them.

Finding 6 ('Cleaning process preserved all images'): 'images.after' isn't in ['cleaning_summary...

+33s Revised output accepted

+35s Revised output accepted

+35s Reviewer requested 1 change(s) and 1 addition(s)

Finding 5 (other): The finding incorrectly identifies 'image' as the target column when the user goal and metadata clearly state that 'label' is the target column. Fix: Update the target leakage analysis to use 'label' as the target column instead of 'image'. Missed: Target leakage was evaluated ...

+37s Reviewer approved the findings

+38s Findings failed the fact check

Finding 1 ('Exact duplicate row removed'): 'duplicate_rows_pct' isn't in ['tbl_overview_001', 'cleaning_summary_001']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them. Finding 2 ('Data quality score improved'): 'quality_before' isn't in ['tbl_quali... ... and 10 more in the HTML report.

Cleaning applied

Each data type was cleaned by its own crew.

Charts

Data quality by type

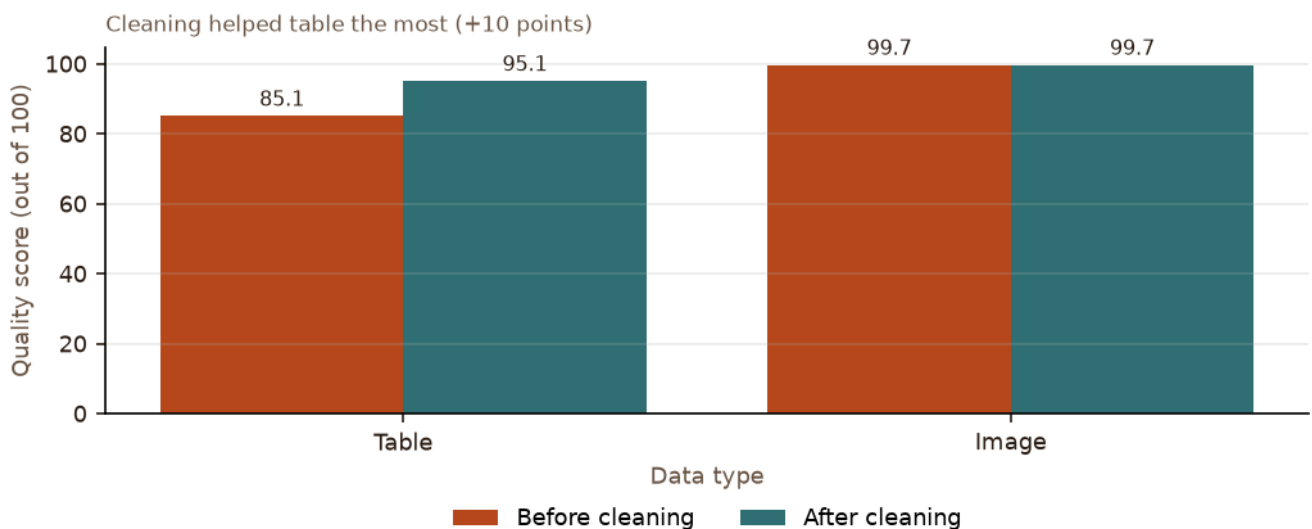


Table: Missing values by column

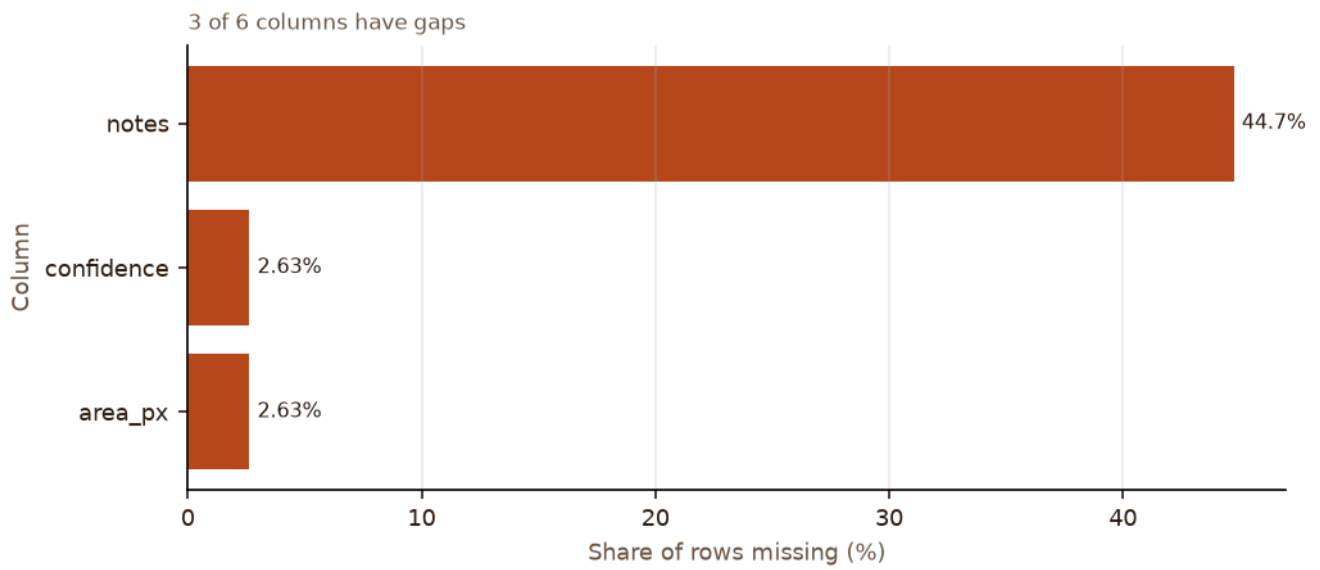


Table: Distribution of confidence

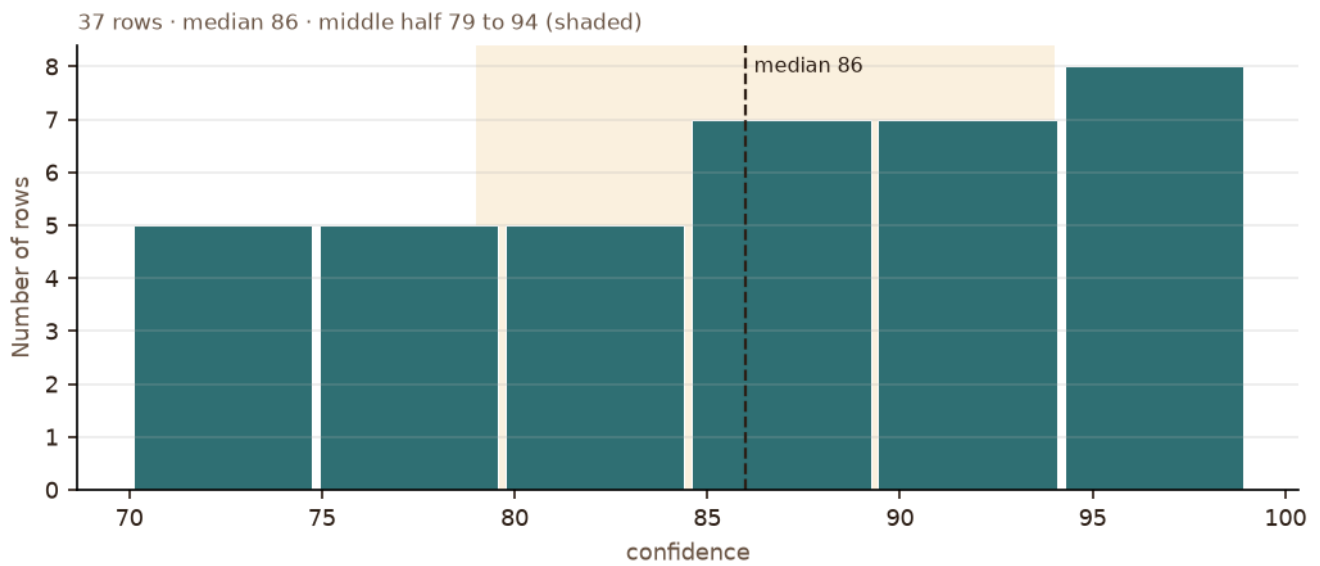


Image: Images per class

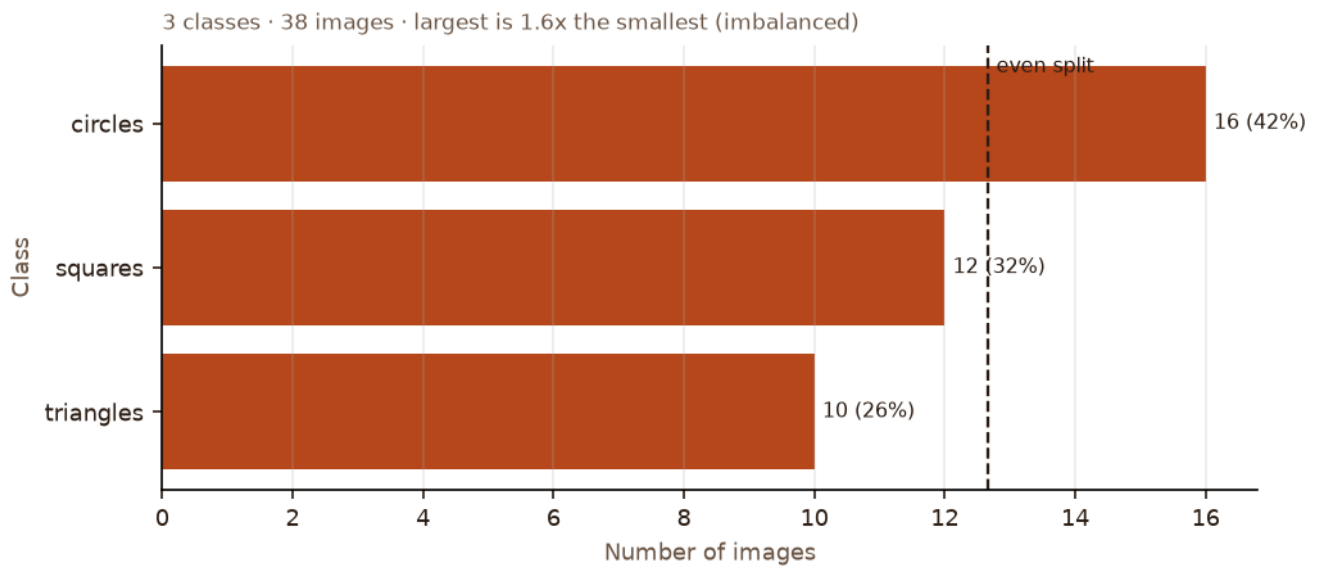
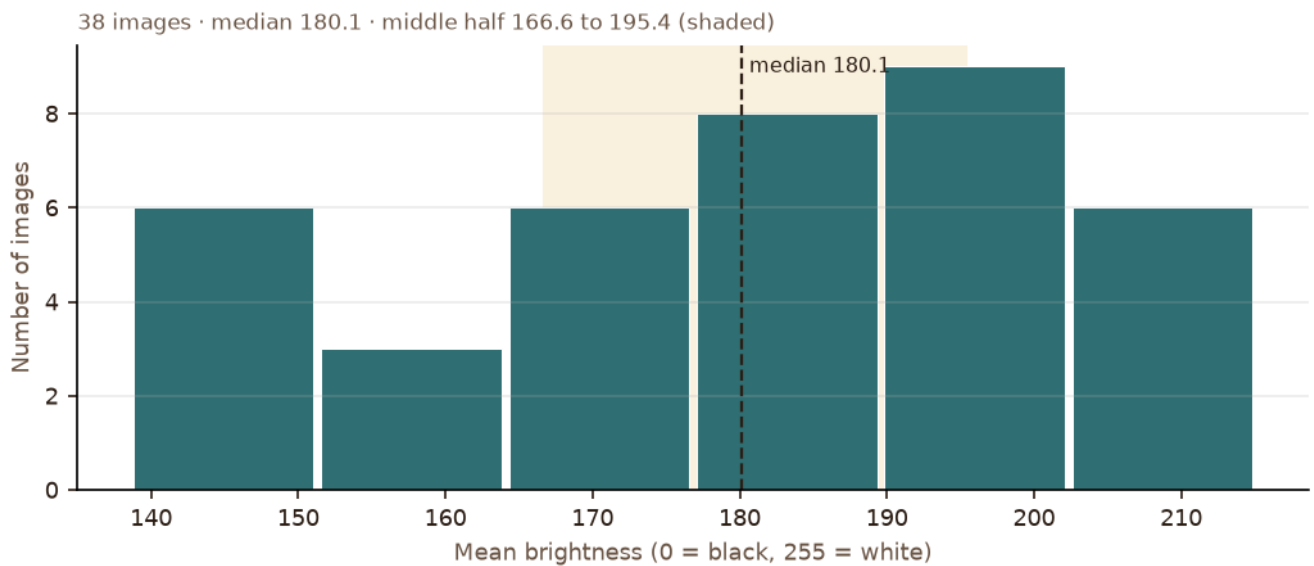


Image: Brightness



Every number in the findings was checked against the evidence artifacts listed. Models used: gemini-3.1-flash-lite, gemini-3.5-flash-lite, gemini-3.8-flash.